



Noice Cancellation- Normalized LMS Algorithm

第10組

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Normalized LMS Algorithm

$$y(n) = w^T(n)x(n) = \sum_{i=0}^{M-1} w_i(n) \cdot x(n-i)$$

$$e(n) = d(n) - y(n)$$

$$w(n+1) = w(n) + \frac{\mu}{\epsilon + \|x(n)\|^2} \cdot e(n) \cdot x(n)$$

$$w_i(n+1) = w_i(n) + \frac{\mu}{\epsilon + \|x(n)\|^2} \cdot e(n) \cdot x(n-i)$$

where $i = 0, 1, 2, \dots, M-1$

M:濾波器階數

μ :步長參數

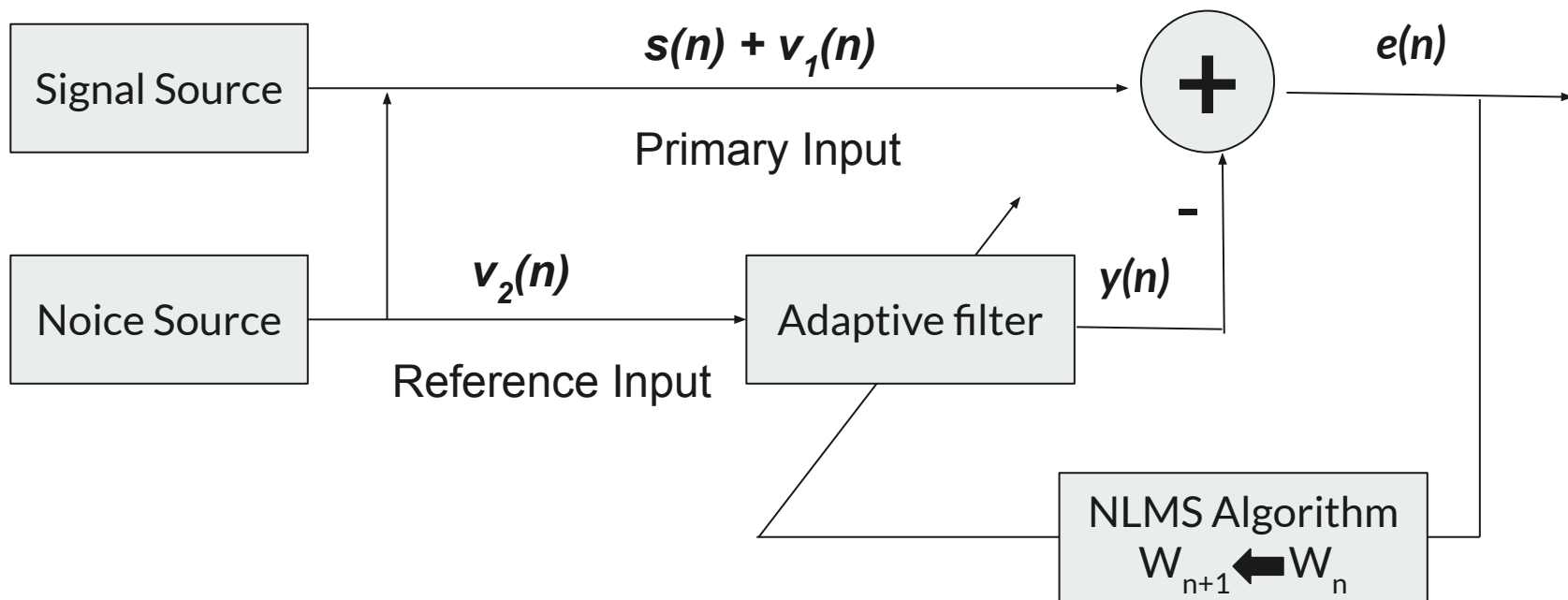
$x(n)$:輸入訊號

$y(n)$:輸出訊號

$d(n)$:理想輸出訊號

ϵ :避免除以零的小常數

Noise Cancellation



Noice Cancellation

$$y(n) = w^T(n)v_2(n) = \sum_{i=0}^{M-1} w_i(n) \cdot v_2(n - i)$$

$$e(n) = d(n) - y(n)$$

$$w(n + 1) = w(n) + \frac{\mu}{\varepsilon + \|v_2(n)\|^2} \cdot e(n) \cdot v_2(n)$$

$$w_i(n + 1) = w_i(n) + \frac{\mu}{\varepsilon + \|v_2(n)\|^2} \cdot e(n) \cdot v_2(n - i)$$

where $i = 0, 1, 2, \dots, M-1$

Noice Cancellation

利用($e(n)$ =訊號源- $y(n)$)剩餘的訊號, 即希望保留的語音

參數調整

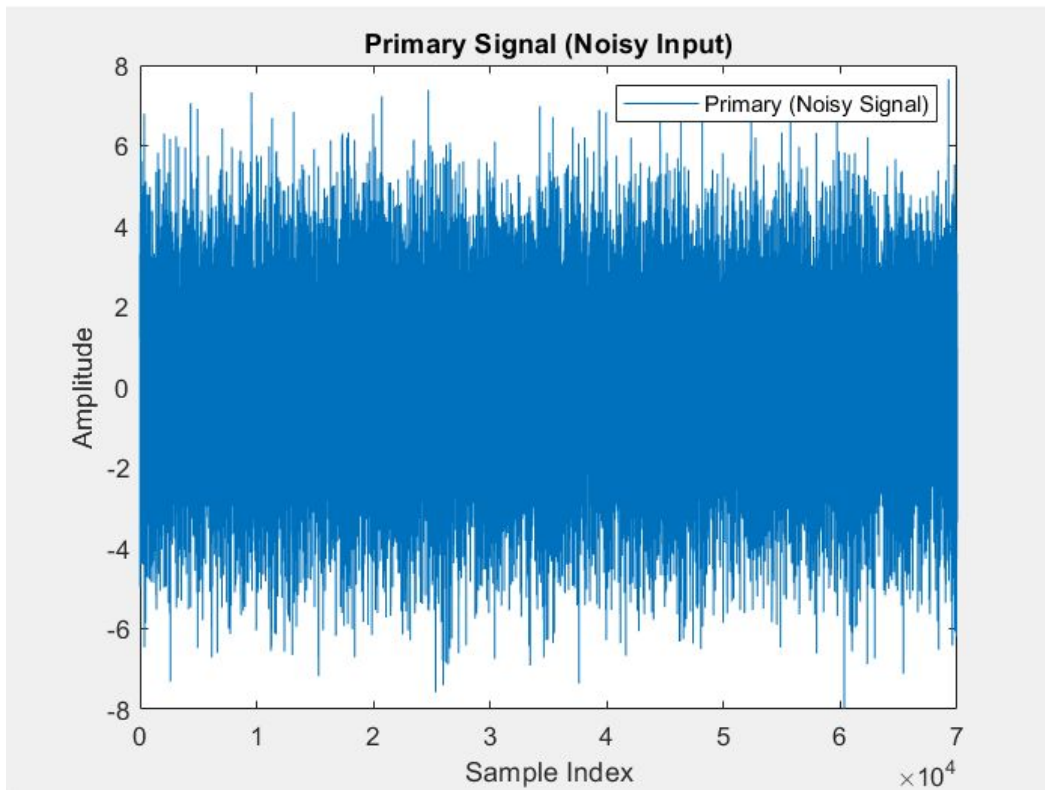
- 濾波器階數(Filter Order): 影響模型的學習能力, 階數高可擬合更複雜噪音
- 步長參數 μ : 太大會發散, 太小則收斂慢

➡ **目的:** 濾波器會在多次迭代後收斂, 也就是誤差 $e(n)$ 穩定下來, 這時可以視為已成功去除大部分噪聲

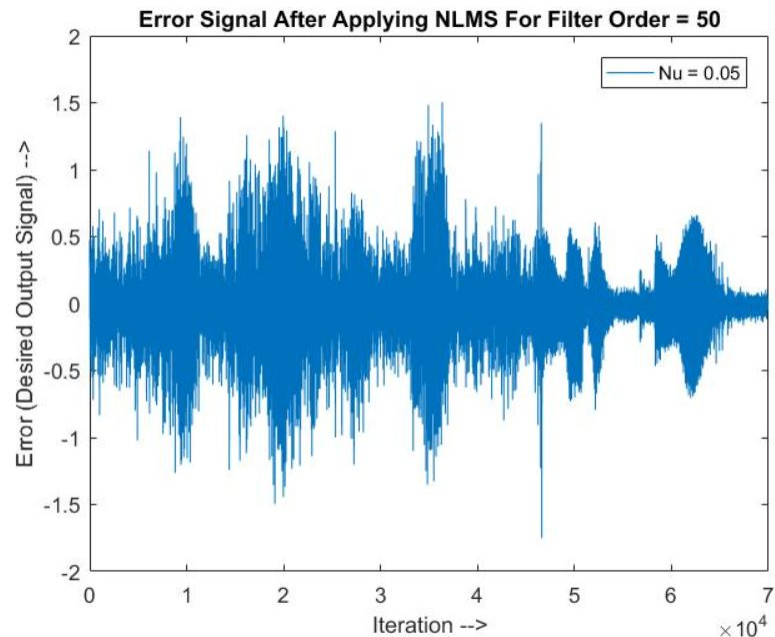
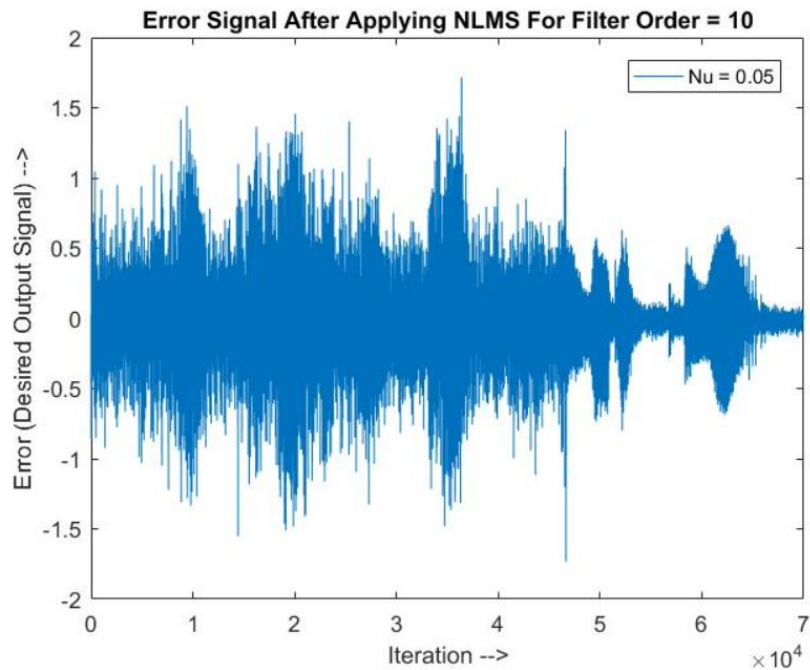


模擬結果

Primary Input



模擬結果



實驗模擬結果

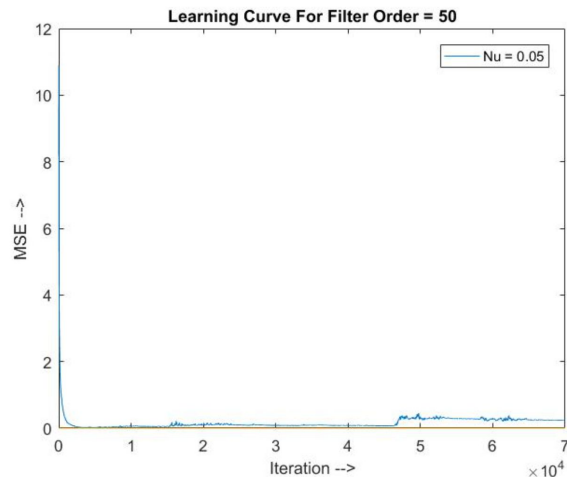
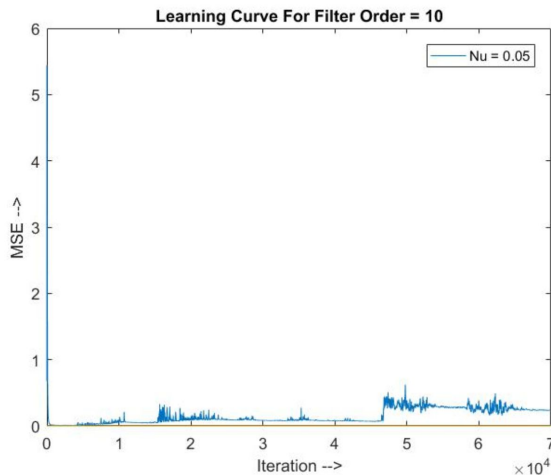
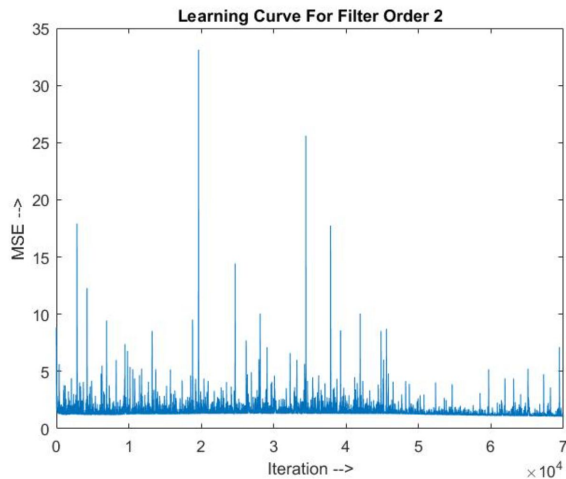


原始聲音 (人聲+噪聲): 

訓練聲音 (噪聲): 

濾波器輸出結果: 

Filter Order (M)多寡對MSE造成的影響



M上升時

優點: 較能夠抵抗雜訊, 更好的還原訊號, 並收斂速度加快。

缺點: 計算複雜度上升。

改變步常參數 (μ)



若將 μ 設太大: 權重更新幅度也相對加大, 收斂速度較快, 但容易震盪使得難以穩定收斂一個值。

若將 μ 設太小: 導致收斂速度緩慢

實驗:

不同 μ 值的MSE比較、不同 μ 值對Tap Weight 造成的影響

預期結果:

μ 值= 0.01 | 穩定但收斂慢

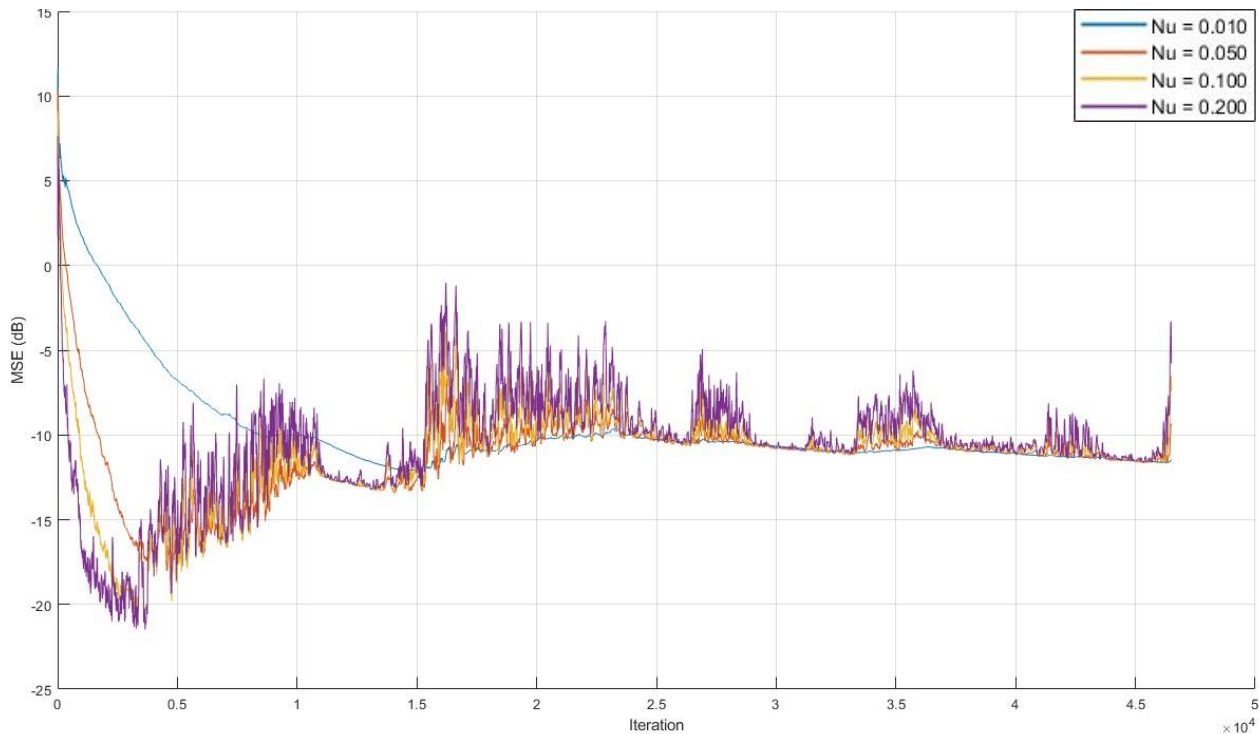
0.05 | 通常為最佳平衡

0.1 | 收斂快但可能有震盪

0.2 | 有機會發散或過擬合

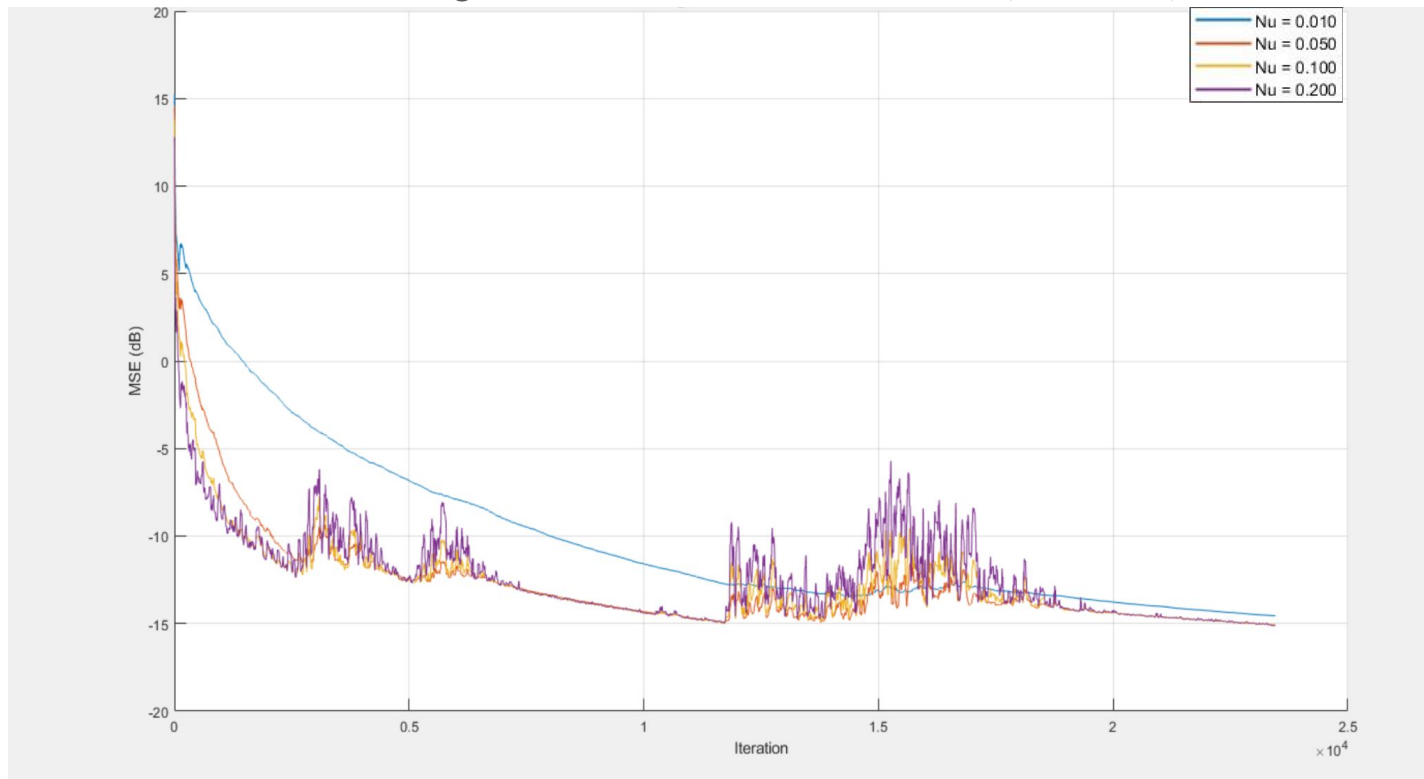
不同 μ 值的MSE比較

MSE Learning Curve before 46.5k for Different Nu(Order=50)



不同 μ 值的MSE比較

MSE Learning Curve after 46.5k for Different Nu(Order=50)



步長參數 μ 對 tap weight 所造成的影響



發現:

M=1: 僅管 μ 設定小值, 也會導致 tap weight 更新權重過快, 使得難以收斂

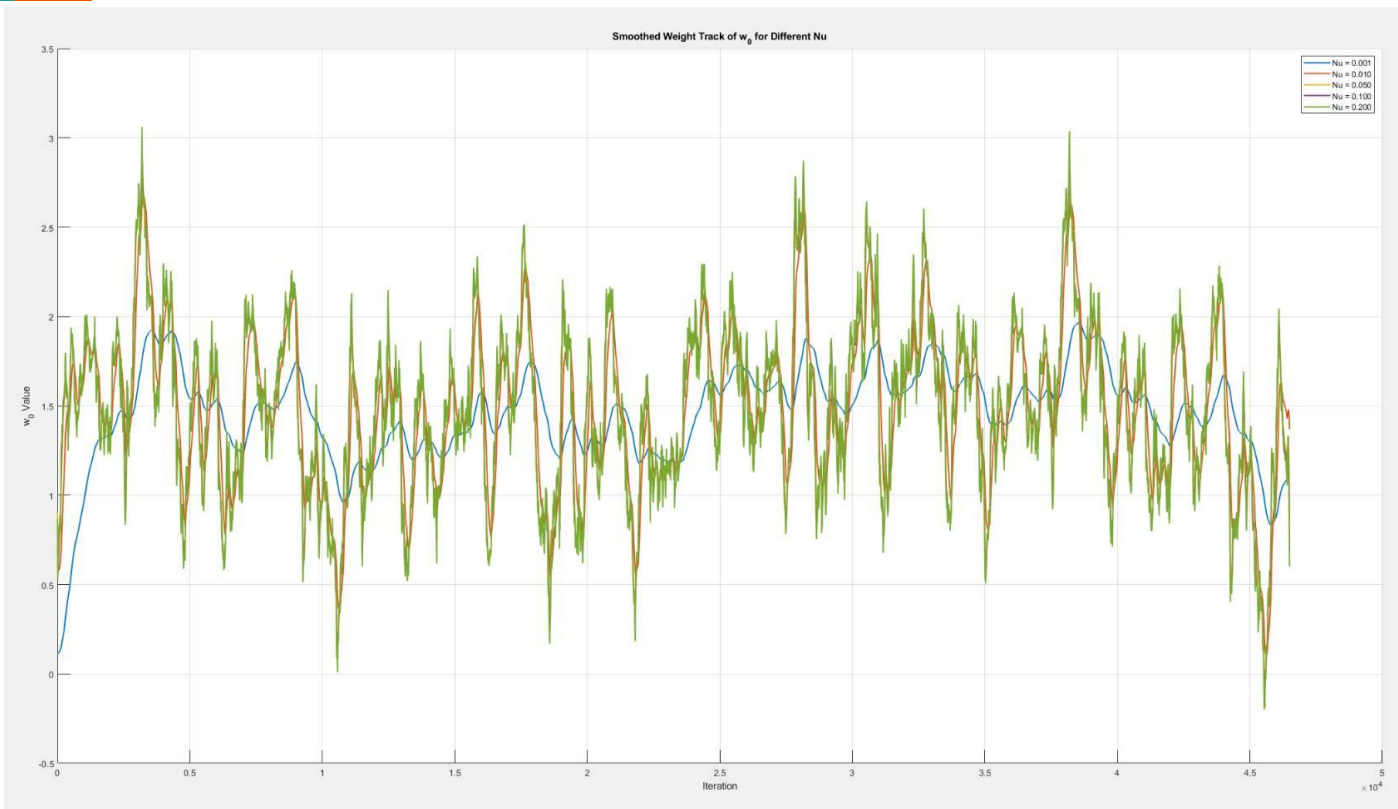
M=50: 恢復正常狀態, 便可觀察 μ 越大, 收斂速度越快且易震盪的情形

結論:

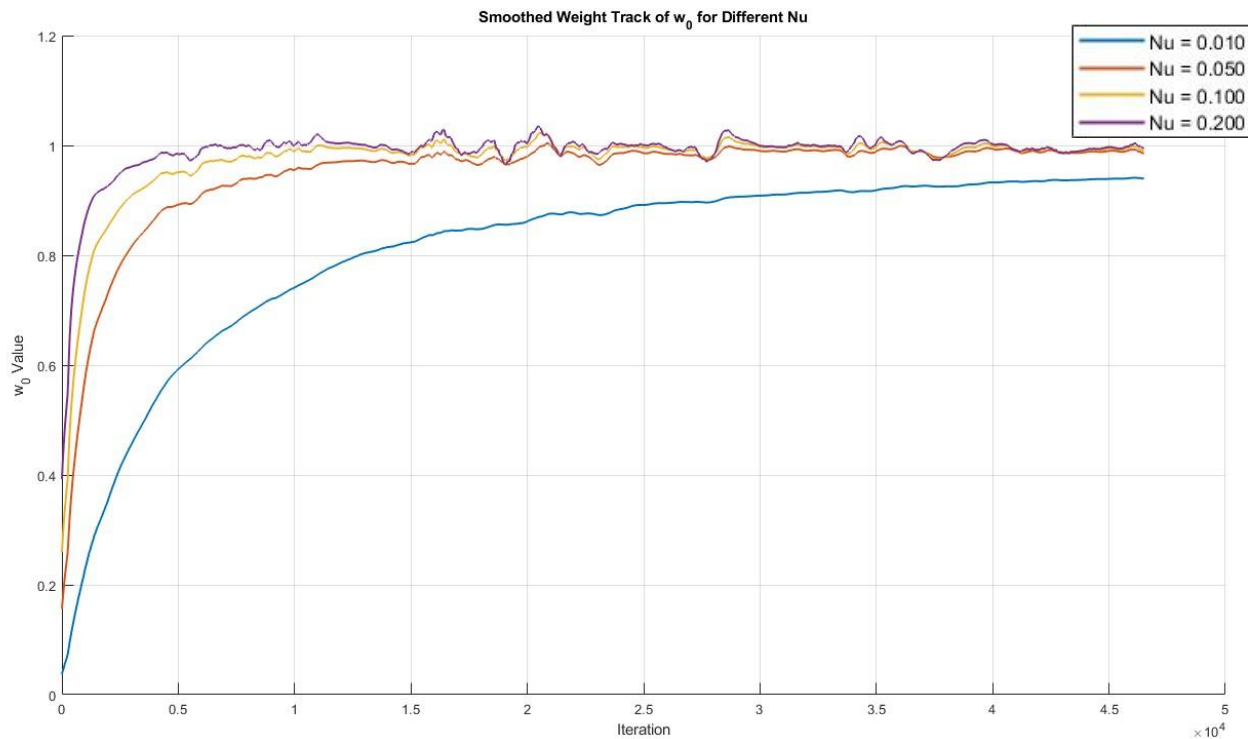
若將 μ 設太大, 則收斂速度較快, 但容易震盪, 較難以收斂穩定一個值。

若將 μ 設太小, 則收斂速度緩慢

With $M=1$, $\mu= 0.01、0.05、0.1、0.2$



With $M=50$, $\mu=0.01, 0.05, 0.1, 0.2$



參考資料



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